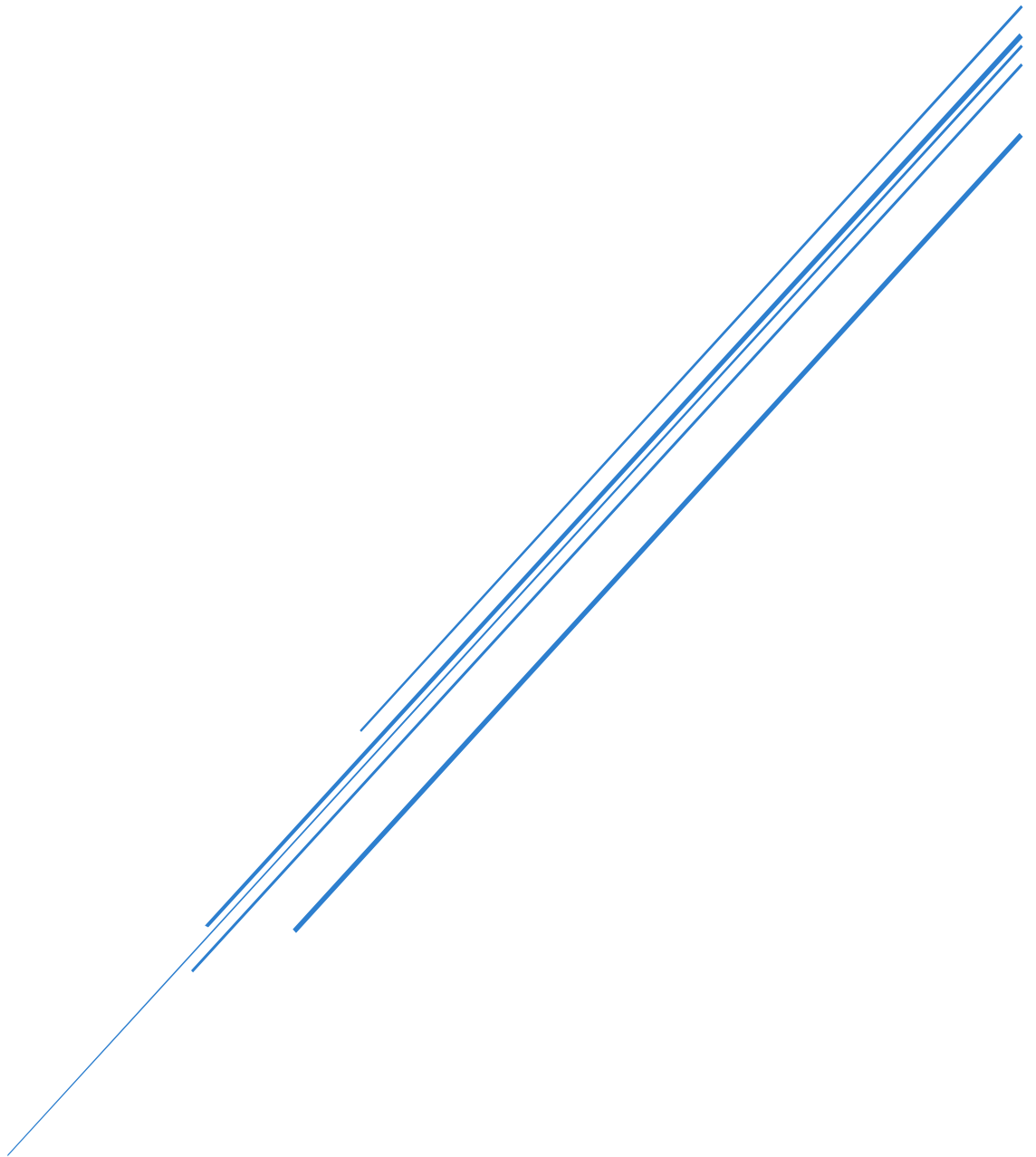


DISSERTATION – INTERIM SUBMISSION

Albert Skalinski



DkIT
07/06/2026

Table of Contents

Background/Introduction	2
Literature Review	4
Exploration of Data and Methods (Artefact)	7
Proposed Future Analysis	14
References	15

Background/Introduction

The popularity of music is a phenomenon that is difficult to explain clearly, as it depends both on the features of a track and the external factors, such as popularity of the artist, promotion, playlist presence, social trends and the way the streaming platforms work. The modern music industry is primarily formed by giant services (such as Spotify), which allow the evaluation and analysis of songs using quantitative data. Due to this fact, music might be explored not only on the cultural level, but also as an object of data analyses. Each track can be described by a set of audio features, such as danceability, energy, loudness, acousticness, instrumentalness, valence, tempo, speechiness, or liveness. These features enable to check whether measurable features that are connected to higher song popularity exist.

The main research problem of this Dissertation is the following: *What musical features drive song popularity?* This problem is important, because the popularity of a track is an artistic, commercial and analytical matter. Understanding the underlying factors for music popularity can help artists and record labels in making decisions about production, promotion and targeting the right audience. From the data analytics perspective, this research problem allows to apply a typical pipeline: data cleaning, feature engineering, Exploratory Data Analysis, data visualisation, model building, model evaluation and the interpretation of feature importance.

This project concentrates on predicting song popularity based on audio features provided by Spotify, but its goal is not only to build a predictive model. Another important element is also the analysis of whether the popularity determinants change over time. It means, that this Dissertation combines two approaches: a predictive and an interpretational one. The study is going to try to answer the question of how precisely song popularity can be predicted based on the provided features. On the other hand, it is going to check whether the features important for popularity stay stable in different periods of time, or their meaning changes with musical trends.

Based on that, the project is based on two main research questions:

1. How have the structural determinants of song popularity evolved over time in contemporary music?
2. How accurately can song popularity be predicted using Spotify audio features?

The first research question involves change over time. It assumes that the song popularity is not static, because the listeners' preferences, production styles and dominant music genres change over decades. Features such as loudness, energy, danceability, or acousticness might have had different meanings for popularity back in the day. The second research question concerns the accuracy of the predictions. In this case, the goal is to check whether Spotify audio features contain enough information to predict song popularity with the use of models such as a baseline regression model or something more sophisticated, e.g., XGBoost.

This dissertation is important in the field of Data Analytics, because it shows how cultural and recreational data can be analysed with the aid of quantitative methods. Music is a subjective matter, but streaming platforms convert that into data that can be modelled, compared, and visualised. At the same time, the project has a practical meaning, as similar approaches can be used in recommendation systems, music marketing, or trend analyses in streaming platforms. It is worth noting that the popularity cannot be explained solely by audio features. It is likely that the model based only on Spotify audio features will have limited accuracy, as it does not contain any information on artist popularity, promotional budget, social media presence, or playlist positioning.

The proposed research process contains multiple stages. First, data from Kaggle has been prepared and cleaned: different .csv files have been joined, missing values have been dealt with, problematic values have been removed/handled, and additional temporal features have been introduced (releaseYear & releaseMonth). Next, Exploratory Data Analysis has been conducted to analyse the variable distribution and relationships between the audio features and popularity. The next stage will be producing predictive models and result evaluation, as well as the analysis of feature importance in different time periods.

The main hypothesis of this Dissertation assumes, that Spotify audio features will partially explain and predict track popularity, but their accuracy will be limited due to the lack of the external factors. Additionally, it is to be assumed that the meaning of specific audio features will change over time. It means, that the popularity determinants are not permanent. This report will present the literature review on the topic, and then it will describe the data, methods of data cleaning, and Exploratory Data Analysis. Finally, the plan of further analysis will be presented at the end.

Literature Review

The four analysed papers have been found significant for this project, because they relate to three connected areas: the usage of audio features in song popularity prediction, the temporal change of determinants, and the limitations of models based solely on audio data. They jointly show that audio features can have predictive value but are not enough for a full explanation for a track success.

The first paper (Interiano et al., 2018) is the closest to the first research question of this dissertation, because it analyses how musical features and popularity changed over a long period of time. The authors studied over 500,000 songs released in Great Britain between 1985 and 2015, comparing tracks present in the top charts with tracks outside these lists (Interiano et al., 2018). The results are important, because they show that popular songs differ from the mediocre ones in terms of musical features. They authors stated, that successful songs were happier, brighter, party-like and danceable, whilst being less sad and less relaxed (Interiano et al., 2018). This suggests that the popularity might be partially connected to measurable features of sound. At the same time Interiano et al. showed that these dependencies are not static. They noticed a general decline of happiness and brightness with the addition of an incline of sadness in one particular period of time (2018). This supports the fact that the popularity determinants should be analysed temporarily, and not only as one global model.

The prediction results from Interiano et al. are also important for the second research question of this Dissertation. The authors used Random Forest to predict whether the song will hit the charts or not (2018). Their result shows that the same musical features had limited accuracy: “using the acoustic characteristics available to us, we were not able to exceed a prediction accuracy of 0.74” (Interiano et al., 2018, p. 14). Only after adding a new variable called “superstar” (former artist popularity) increased the prediction accuracy to ~0.86 (Interiano et al., 2018). This shows that audio features are important, but popularity is also strongly dependent on external factors, such as artist popularity. For this project it means, that a model based solely on Spotify audio features might achieve mediocre accuracy, so its results should be evaluated with caution.

The second paper (Saragih, 2023) concentrates directly on Spotify audio features. The study had been conducted on Kaggle data (over 90,000 observations) and regarded the song popularity in Indonesian people (Saragih, 2023). The author compared many classification and regression models. The best classification model turned out to be Random Forest Classifier, which achieved ~0.697 in accuracy and ~0.694 in F1-score (Saragih, 2023). The best regression model was Extra Trees Regressor with the score of 0.6857 in R^2 and 12.33 in RMSE (Saragih, 2023). The author states directly: “The study indicates that both machine learning models yield a predictive power of nearly 69% by using classification and regression algorithms.” (Saragih, 2023, p. 705). This suggests that Spotify audio features can be used for predicting popularity, but their predictive power is mediocre, rather than strong.

The key takeaway from this paper is the analysis of feature importance. The author described ten most important features for classifying popularity: release date, loudness, instrumentalness, acousticness, duration, speechiness, energy, valence, danceability, and liveness (Saragih, 2023). A lot of the same features are present in the dataset used in this Dissertation. Saragih suggests, that highly popular songs in Indonesia were usually newer, louder, less acoustic, less instrumental, less speech, more energetic, and usually recorded in a studio (2023). At the same time, the study has certain limitations. It limits itself to the Indonesian market and its Spotify users, so the results might be biased towards that local culture and the characteristics of the platform (Saragih, 2023). In the context of this project, this paper is useful because it is an example of how to connect regression and feature importance.

Petitbon and Hitchcock expand the context by not focusing on individual audio features, but on genre popularity changes in time (2022). The authors analysed the data from Billboard Hot 100 and Village Voice Pazz & Pop from 1974 to 2018 using multinomial logit models to model changes in genre preferences (Petitbon, Hitchcock, 2022). The results show that music popularity is strongly dependent on the time period and broader cultural changes. For example, the authors state that “Rock steadily declined in the late 1980s through 2000s” (2022, p. 177). It shows that popularity might change not only because of individual audio features, but also on the genre level.

Petitbon and Hitchcock’s results are also significant for interpretation of temporal results. If musical genres change their popularity over time, then one can expect that the audio features connected to these genres will also change. For example, the increase of popularity of Rap/Hip-Hop or Electronic music might affect the significance of speechiness, energy, loudness, or danceability. This supports the decision of this analysis to include feature importance in different time periods.

Finally, Zangerle et al. (2019) concentrate on hit song prediction whilst using low-level and high-level audio features and deep learning. The authors used Million Song Dataset and defined a hit song as a track, which appeared in Billboard Hot 100 (Zangerle et al., 2019). Their approach was based on connecting low-level features, such as spectral, rhythm and tonal descriptors with high-level features, such as mood, genre, voice, or release year (Zangerle et al., 2019). The best results were achieved when the model connected both types of features, which suggests that different levels of music representation tie-in with each other (Zangerle et al., 2019). This shows that the prediction might be the most accurate, whenever one does not limit himself to one group of variables.

The authors stated that “the release year information is the most important high-level feature” (2019, p. 324). It confirms that time should not be treated as a technical variable, but as an important feature crucial in this context. If release year significantly improves prediction performance, it means that the popularity of a track will be highly dependent on the release date and the dominant trends at that time.

Comparing all four papers, a couple of conclusions can be made. Firstly, audio features have an actual value in predicting popularity but cannot fully explain the success of a track. Interiano et al. show that adding artist information increases the accuracy (2018), whilst Saragih and Zangerle et al. show that models based on audio features achieve moderate, but useful results (2023) (2019). Secondly, time is an important factor. Interiano et al., Petitbon and Hitchcock, and Zangerle et al. prove that music and people's preferences change over time (2018) (2022) (2019). Thirdly, feature importance is as important as model accuracy, because the goal of the project is not only to predict, but also to interpret.

Based on all of the above, the literature shows a clear research gap. Many studies focus on prediction accuracy or on general musical trends, but connecting both perspectives gets less attention. Therefore, this Dissertation will address that research gap through analysing Spotify audio features, modelling, and comparing feature importance in different periods. This way the project is not limited to answering the question of whether the popularity can be predicted. It also studies whether the features responsible for popularity change with the evolution of contemporary music.

Exploration of Data and Methods (Artefact)

The data used in this project comes from a publicly available Kaggle dataset containing information on songs and their Spotify audio features. The dataset consists of two main groups of variables: descriptive features and audio features. Descriptive features consist of the track name, track artist, track album name, and track album release date. It also contains different IDs, such as track ID, track album ID, and playlist ID, which allow to distinguish between songs, albums, or playlists. Additionally, there are variables that describe the context of the given playlist, such as name, genre, or subgenre. The most important target variable is popularity, which is a score between 0 and 100 calculated based on the popularity of the track in relation to other songs.

The second group consists of Spotify audio features, which describe measurable properties of tracks. Energy measures the intensity and activity, where energetic tracks are usually fast, loud, and dynamic. Tempo is measured in beats per minute. Danceability describes how suitable the track is based on factors such as tempo, rhythm, stability, beat strength, and overall regularity. Loudness states the loudness in decibels, whilst liveness is the probability of the track being recorded live. Valence measures the positivity of the track – high valence means a happier sound, whilst low valence means a sadder or angrier sound. Speechiness measures the presence of spoken words, and instrumentalness calculates the probability of the track not having vocals. Mode describes the modality of the song, whilst key means is represented as an integer (0-11). The dataset also contains duration of the track in milliseconds and acousticness, which is a confidence measure stating whether the song is acoustic or not.

This dissertation does not involve a contact with participants, and it does not collect any personal data. The data comes from a publicly available dataset and contain song information, not private user behaviour. Spotify accounts, listening history and data that could be tied to a person are not analysed. Additionally, it is worth noting that track popularity is a platform measure, and not a universal measure for the significance of the song. The results of this Dissertation should be analysed within the closed Spotify framework, and not as a universal explanation of a musical success.

The first step of this project was data cleaning. Two original .csv files were loaded. The structure, the number of observations and the available columns were inspected. Then, both .csv files were joint into one dataset. The initial inspection included checking data types, missing values, duplicates and the quality of release dates. Track album release date was thoroughly inspected, as it is crucial for the temporal analysis conducted in this Dissertation. The dates were converted into a datetime format, and they were further broken down into release year and release month. Additionally, 125 missing/incomplete dates were detected, which were also cleaned up. After conducting all these steps, the dataset was saved to a new .csv file and used in further steps.

The second stage was feature engineering. Release year and release month were created, as stated above. On top of that, popularity group was also created. Due to this fact, track popularity could be interpreted as a categorical value in addition to its base numerical nature. This later made it easier to compare various audio features to popularity groups. A similar approach is used by Saragih, where he creates a variable called popularity class, diving it in a similar manner (2023).

Furthermore, an Exploratory Data Analysis was conducted. The main goals were the following: understanding the structure of the data, evaluation of variable distribution, and analysis of relationships and correlations between variables. Different Python libraries were used, such as pandas, numpy, matplotlib, and seaborn. Basic information on the dataset was inspected, such as the number of observations, column types and descriptive statistics. Later, the distributions of track popularity, release year and audio features were analysed.

The visualisations showed a couple of important insights. Firstly, the dataset is skewed towards recent music, which means that older songs are less represented compared to the more recent ones. It is an important limitation, because older songs that are still popular on Spotify might be extreme cases and do not represent all the older songs as a whole. Secondly, the EDA did not show any simple and significant relationships between the majority of individual audio features and popularity. It means, that popularity is probably a multivariate phenomenon that cannot be explained in one simple variable, such as energy, danceability, or tempo.

The strongest positive correlation occurred between loudness and energy, which is intuitive, because energetic tracks are usually louder and more intensive. Moderate positive correlations occurred between danceability and energy, danceability and loudness, and also between Valence and a few other features. A strong negative correlation occurred between acousticness and energy, whilst moderate negative correlations could have been seen in energy and instrumentalness, loudness and instrumentalness and loudness and acousticness. In relation to popularity, a semi-moderate negative correlation could have been seen with release year, instrumentalness and acousticness. These results show that highly popular tracks in this dataset are usually less instrumental and not so acoustic.

The developed artefact can be defined as a temporal song-popularity prediction and interpretation pipeline. It is not a single model, but a whole analytical process: cleaning the dataset, feature engineering, EDA, and future modelling and feature importance interpretation. The planned pipeline includes the following: data preparation, time period division, the baseline regression model building, a more advanced model application, metrics evaluation, and, finally, feature importance comparison between time periods. Such artefact answers both research questions, because it will allow to evaluate the accuracy of the predictions and check how temporal the musical features are (figure 1).

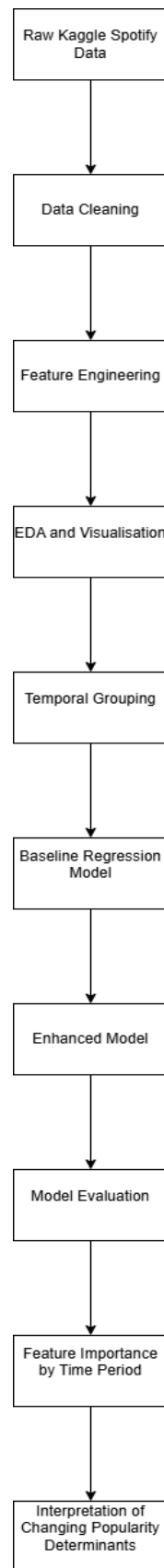


Figure 1 - Artefact Development Flowchart

The snippet below (figure 2) shows how the artefact will choose audio features, define the target variable, create temporal groups based on the release year and prepare the data for the modelling stage.

```
1 features = [  
2     "danceability", "energy", "loudness", "speechiness",  
3     "acousticness", "instrumentalness", "liveness",  
4     "valence", "tempo", "duration_ms", "release_year"  
5 ]  
6  
7 target = "track_popularity"  
8  
9 x = clean_data[features]  
10 y = clean_data[target]  
11  
12 clean_data["period"] = pd.cut(  
13     clean_data["release_year"],  
14     bins=[1950, 1989, 1999, 2009, 2019, 2026],  
15     labels=["pre_1990", "1990s", "2000s", "2010s", "2020s"]  
16 )  
17  
18 baseline_model = LinearRegression()  
19 enhanced_model = XGBRegressor()  
20  
21 # compare model performance and feature importance by period
```

Figure 2 - Prototype Design Snippet

Artefact design visuals were prepared in the EDA notebook of the project. They show how a cleaned dataset can be transformed into clear analytical outputs (figure 3) (figure 4) (figure 5) (figure 6).

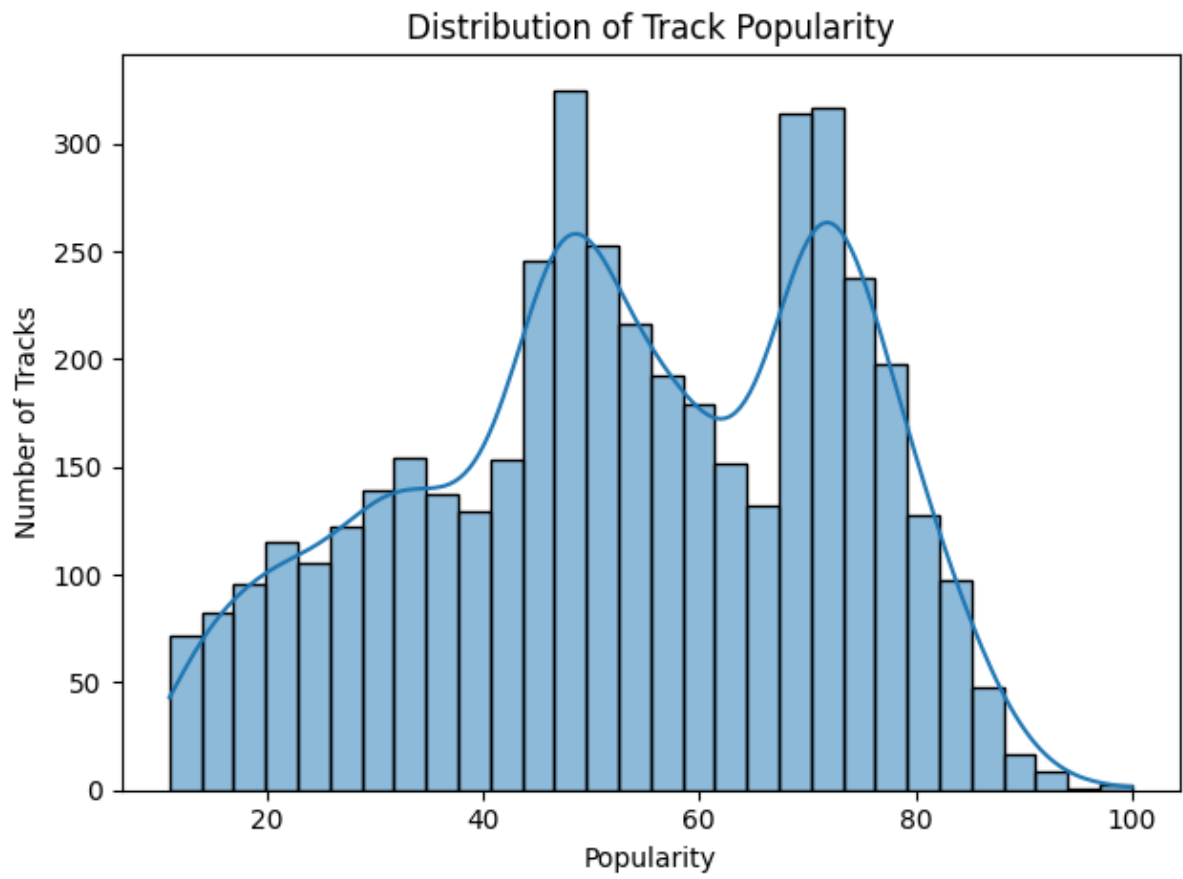


Figure 3 - Track Popularity Distribution

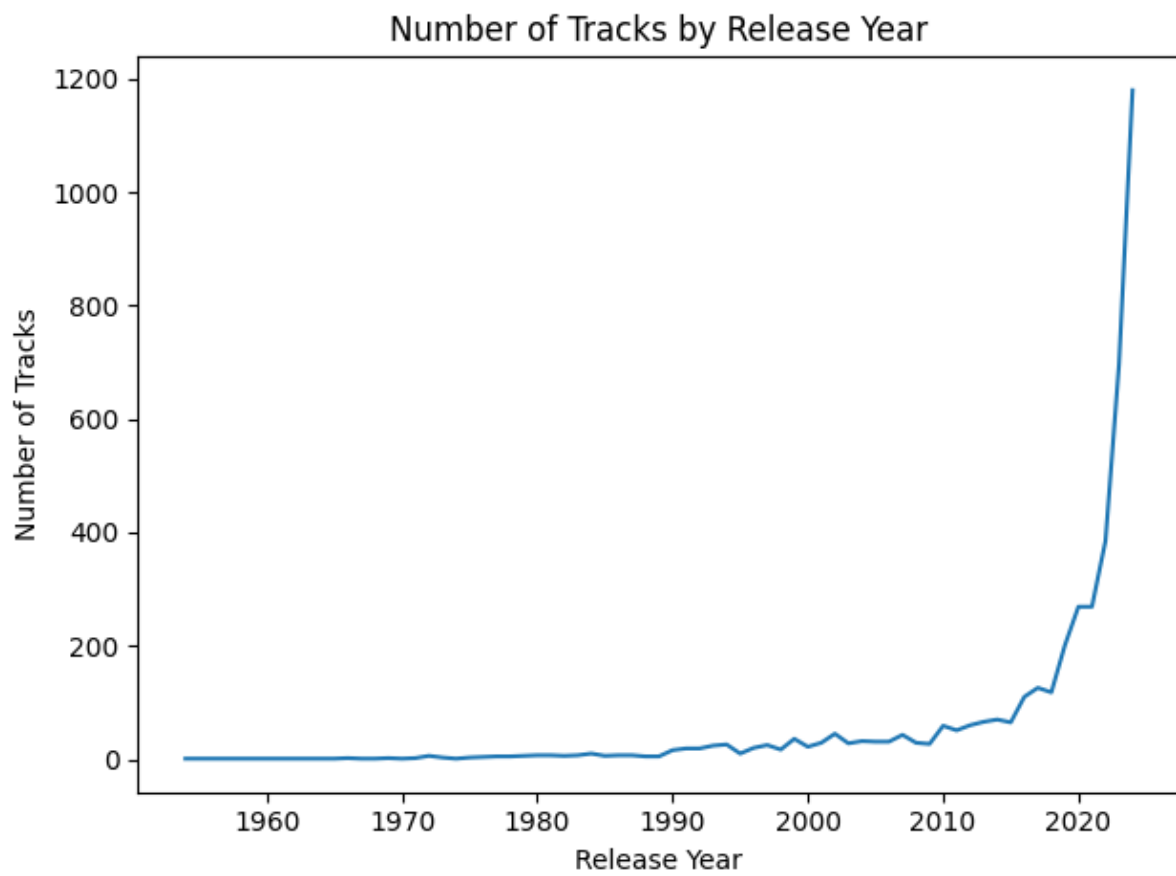


Figure 4 - Number of Tracks by Release Year

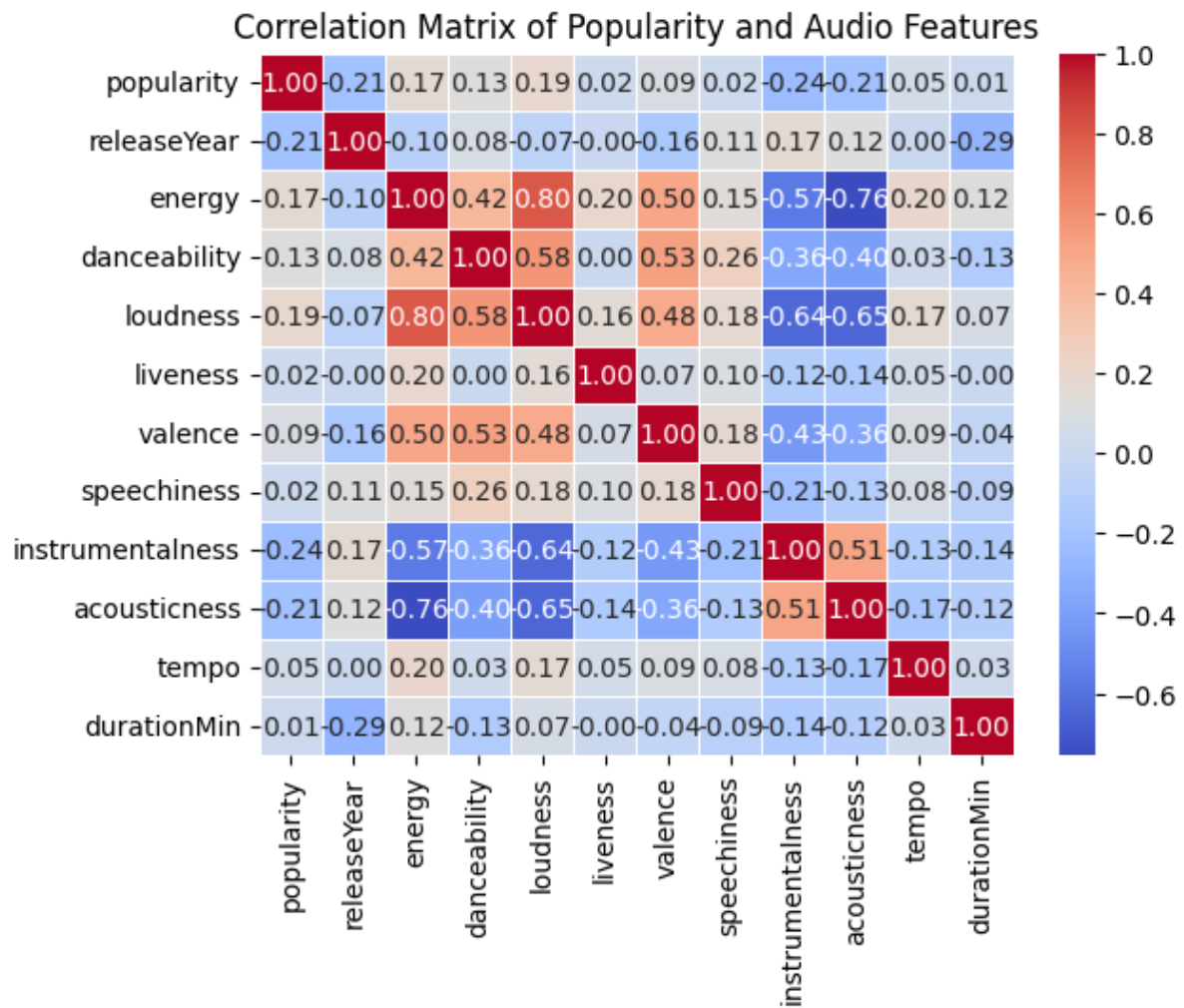


Figure 5 - Correlation Heatmap

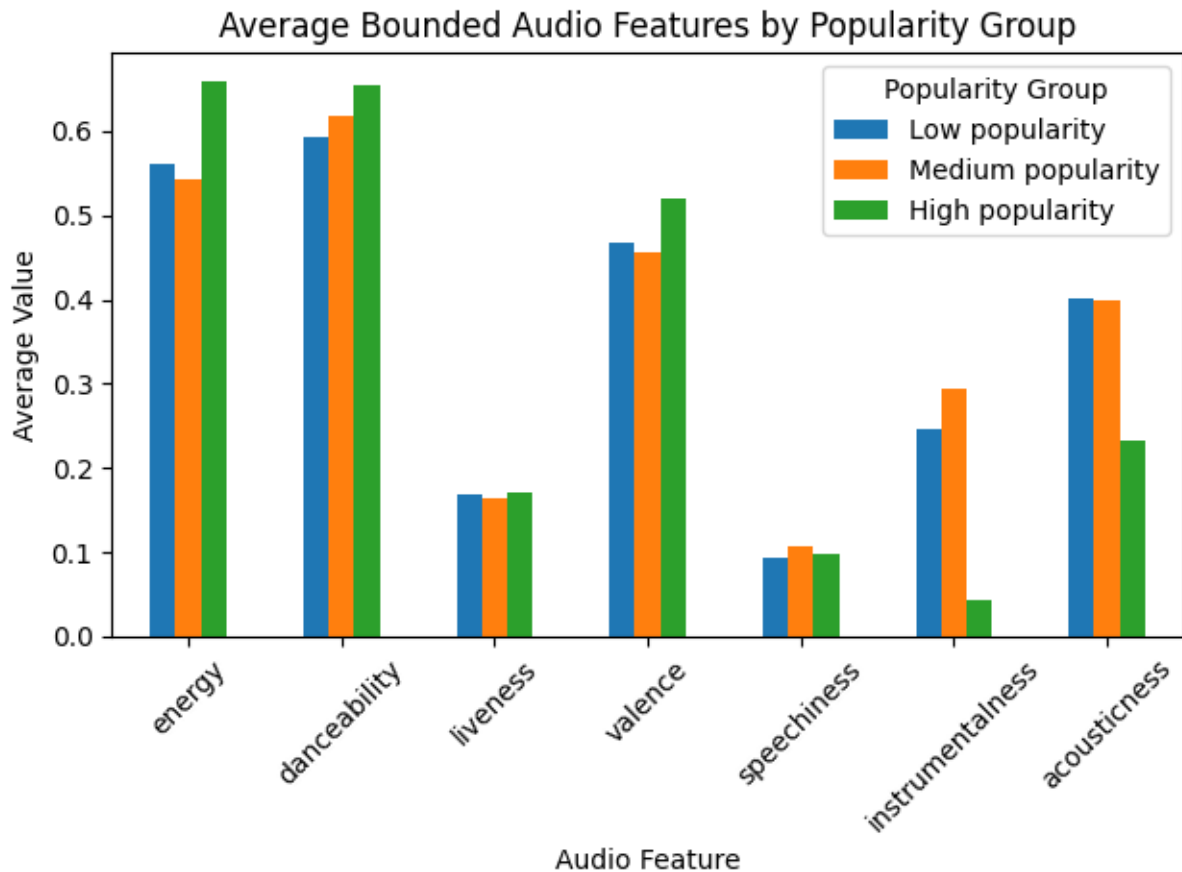


Figure 6 - Audio Features by Popularity Group

Proposed Future Analysis

The next stage of the project after the EDA will be predictive modelling. The data will be grouped into temporal groups so it can be compared to the popularity determinant in various time periods. This stage is directly related to the first research question, as it will evaluate the meaning of features such as energy, loudness, danceability, acousticness, or instrumentalness over time.

Next, a baseline regression model will be built, which will be a starting point for a more sophisticated and advanced method. Popularity will be treated as the target variable and evaluation metrics, such as MAE, RMSE, or R^2 will be used. In the next step, an advanced model will be implemented (e.g., XGBoost) to check whether it will improve the predictive performance or not.

Lastly, feature importance analysis will be the last step. The models will be evaluated not only in terms of accuracy, but also in terms of interpretability. It will be crucial to compare the feature importance between different temporal groups, as it will answer the second research question. The final artefact will then be enhanced with more visuals, such as Actual vs Predicted Popularity, or Feature Importance by Time Period. The results will be described in the context of limitations of the Dissertation, especially the fact that audio features do not include external factors, such as artist popularity, or social media trends.

References

Interiano, M., Kazemi, K., Wang, L., Yang, J., Yu, Z. and Komarova, N.L. (2018). Musical trends and predictability of success in contemporary songs in and out of the top charts. *Royal Society Open Science*, [online] 5(5). <https://doi.org/10.1098/rsos.171274>

Harriman Samuel Saragih (2023). Predicting song popularity based on spotify's audio features: insights from the Indonesian streaming users. *Journal of Management Analytics*, 10(4), pp.693–709. <https://doi.org/10.1080/23270012.2023.2239824>

Petitbon, A.M. and Hitchcock, D.B. (2022). What Kind of Music Do You Like? A Statistical Analysis of Music Genre Popularity Over Time. *Journal of Data Science*, 20(2), pp.168–187. <https://doi.org/10.6339/22-jds1040>

Zangerle, E., Huber, R., Vötter, M. and Yang, Y.-H. (2019). HIT SONG PREDICTION: LEVERAGING LOW-AND HIGH-LEVEL AUDIO FEATURES. [online] <https://doi.org/10.5281/zenodo.3258042>

The above references have been generated with the aid of this tool:

<https://www.mybib.com/tools/harvard-referencing-generator>

Dataset used: <https://www.kaggle.com/datasets/solomonameh/spotify-music-dataset/data>

The code used in this submission was generated with the aid of [AI](#).